

AI Applications Lecture 4

Licenses and Openness

SUZUKI, Atsushi^a

^aI thank my collaborator, Jing WANG, for cooperating in creating the teaching materials.

Outline

1. Introduction
2. What Does "Open" Mean?
3. What rights do we have on open models? - License
4. Model Versions and License Differences Across Versions
5. Summary
6. Next Time

Review of the previous lecture

In the last lecture, we defined the function represented by a neural network as **generally as possible**.

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- A **checkpoint** contains the **specific parameter values** obtained through training.
- Only with both the architecture $f_{(.)}$ and the checkpoint θ is a **specific function** f_{θ} uniquely determined.

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For an input space $\mathcal{X}$, an output space $\mathcal{Y}$, and a parameter space $\Theta \subset \mathbb{R}^D$:

$$\text{Architecture } f_{(\cdot)} : \Theta \rightarrow \{\mathcal{X} \rightarrow \mathcal{Y}\}, \quad \boldsymbol{\theta} \in \Theta \mapsto f_{\boldsymbol{\theta}} : \mathcal{X} \rightarrow \mathcal{Y}.$$

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- **Training** is determining parameters θ^* based on data.
- **Inference** is applying the final function f_{θ^*} .

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- **Open:** Architecture, checkpoints, and even the training code or "recipes" are released.
- A model's **license** determines if others can freely use it, regardless of how "open" it is.

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Example (The Staged Release of GPT-2)

When OpenAI's GPT-2 was announced in 2019, the weights for the largest model were subject to a **staged release** due to safety concerns [1].

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When OpenAI's GPT-2 was announced in 2019, the weights for the largest model were subject to a **staged release** due to safety concerns [1].

- Initially, the most powerful 1.5B parameter model was completely closed.
- The full model was eventually released in November 2019 after a period of study and debate [2].

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Example (LLMs as a Service)

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Example (LLMs as a Service)

As of August 2026, OpenAI provides **ChatGPT-5.6** via its ChatGPT interface and API.

- **Inference is executed on their servers.**
- The **details of the architecture and checkpoints are generally not disclosed** [3].
- Users only interact with prompts and outputs.

From completely closed AI to open AI

Example ("Open" Models and Distribution Platforms)

Models like **Llama** (Meta), **Qwen** (Alibaba), **Mistral** (Mistral AI), and **DeepSeek** (DeepSeek AI) are known as **open (Open Weights/Open Model)**.

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- The **Hugging Face Hub** is a major platform for their distribution [4].
- You can download the architecture and checkpoints to run locally.
- Licenses are (usually) clearly stated in the model repositories. [5, 6, 7, 8, 9]

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By the end of this lecture, students should be able to:

- Explain the various stages of openness of neural network models.
- Avoid the risk of not complying with the license.
- Determine whether a model provided under an major open source software (OSS) license can be used for their own purposes.

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- The goal is to **reasonably prevent** legal risks.
- **Final decisions should be made at your own responsibility. Refer to official documentation [10, 11, 12].**

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Question: When we say an AI model is open, what does exactly mean?

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- The **architecture** (as configuration files or code).
- The trained **checkpoint** (the weights).

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- The **training algorithm** (including the complete training recipe and scripts for reproduction).

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With the architecture $f_{(\cdot)}$ and checkpoint θ , **inference** is possible.

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Remark (Feasibility of Inference and Training)

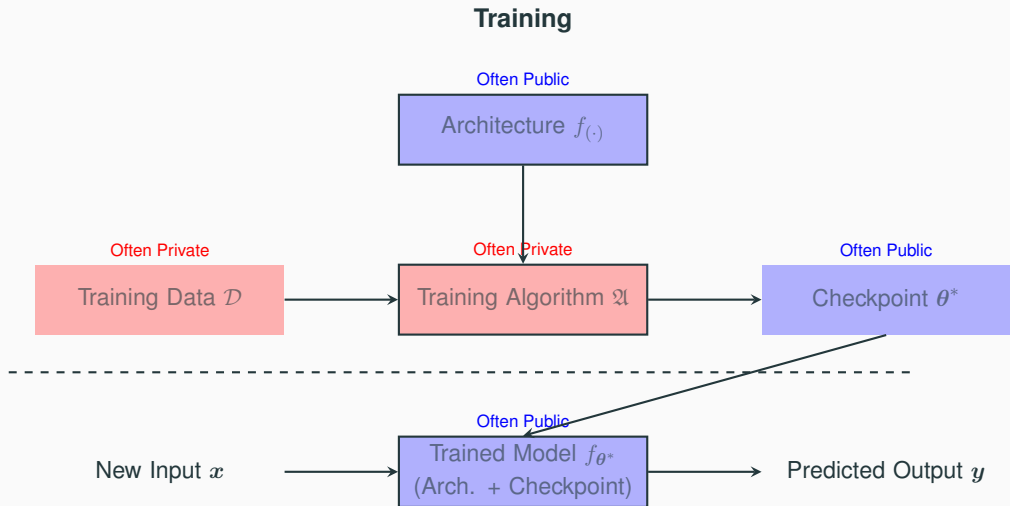
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However, **reproducing** the training (arriving at the exact same θ) is usually impossible unless the data, recipe, random seeds, etc., are identical.

Even with "open weights," if the **insights from the training process** are not disclosed, the reproducibility and potential for adaptation are limited.

Visualizing the relationship

This diagram uses our framework to show what's typically open vs. private.



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- The **weights and detailed implementation remain private**.
- Example: OpenAI's generally available models like GPT-4 and beyond [3].

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Architecture + Checkpoint Release

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- Weights are distributed as **Open Weights**.
- This allows for local execution or use as a starting point for retraining.
- Examples: Mistral 7B (Apache 2.0) [7], DeepSeek-R1 (MIT) [8].
- The **training data and complete training procedures** are often not disclosed.

Specific examples of release stages

Release of Training Code and Recipes

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- Example: AI2's OLMo project [13].

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Question: When we say an AI model is open, what does exactly mean?

Answer: In most cases, it means that its checkpoint (weights) is open (publicly available).

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In particular, each component may be protected by copyright even without a written document.

Hence, even if an AI model is open, using it without permission may have a legal risk.

The distributor of an open AI model may provide permission through a **license**. Hence, understanding licenses is necessary when using open AI models.

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For example, within the same Transformers framework:

- **DeepSeek-R1** weights are MIT [8].
- The **Mistral** family is Apache 2.0 [7].
- The **Llama** family has its own custom license [5, 6].

Architecture vs. checkpoint licenses

Moreover, different licenses may apply to different components. Specifically, the license for the **code** and the license for the **weights** can be different.

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According to the model card, the DeepSeek-R1 checkpoint is licensed under the MIT license [8], even though we can use it via the Hugging Face Transformers library, which is licensed under the Apache 2.0 license [14].

Hence, it is necessary to read licenses for each component of the AI model you want to use.

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Hence, it is necessary to read licenses for each component of the AI model you want to use.

Fortunately, many famous open models are licensed under a major open source software license, so it is beneficial to know those licenses in advance.

Is a checkpoint protected by copyright?

Remark

In most regions of the world, court decisions are limited in whether a checkpoint is protected by copyright. In any case, it is safer to use an AI model only when the license gives explicit permission.

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Remark (Disclaimer)

Final determination of "can I use this?" must be based on the **specific license text and its scope**.

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- **Model/Library example:** The classic machine learning library **scikit-learn** [16].

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- **Model Example**: **YOLOv5** uses AGPL-3.0 [19].

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- Careful consideration is required, especially for commercial use.

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- Such materials should be avoided.

Model-specific licenses (open-weight type)

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The **Community License** for the Llama 3 series allows **commercial use with restrictions** [5, 20].

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- Later versions, like Qwen 3, seem to have standardized on Apache 2.0.

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It also requires that when distributing **derivative works**, the **same use restrictions must be imposed on downstream users**.

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- Key variations include:
 - **CC BY 4.0** (Attribution) [22]
 - **CC BY-SA 4.0** (ShareAlike) [23]
 - **CC BY-NC 4.0** (NonCommercial) [24]

Practical guidelines and caveats

- Seek legal experts' advice.
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- **The legal nature of checkpoints is an immature area:** It is reasonable to consider them as having aspects of both **source code** and **data**. **No legal consensus has been established** [25, 26].
- Seek legal experts' advice.

Practical guidelines and caveats

- Seek legal experts' advice.
- **"Open" does not mean free to use:** The **license text** governs everything. Always check the **LICENSE** and **Terms of Service** on the model page.
- **Apache 2.0 is a widely accepted choice for distributors:** If you want your model to be widely used but have little commercial intent, Apache 2.0 is easy to handle due to the clarity of its **patent clauses** [17].
- **The legal nature of checkpoints is an immature area:** It is reasonable to consider them as having aspects of both **source code** and **data**. **No legal consensus has been established** [25, 26].
- A policy of **"when in doubt, do not use"** is the safest approach.
- Seek legal experts' advice.

License - answer

Question: Under which conditions can we use open AI models?

Answer: Users' rights are limited owing to patents, copyright, contracts, etc. The distributor may give permission through its license.

Outline

1. Introduction

2. What Does "Open" Mean?

3. What rights do we have on open models? - License

4. Model Versions and License Differences Across Versions

5. Summary

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Model versions and license differences

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Example (Qwen 2.5 (Alibaba))

As we saw, different models in the Qwen 2.5 series had different licenses (custom vs. Apache 2.0) [6]. You must check the license for the specific model you wish to use.

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- **Scale:** The number of parameters (e.g., 7B vs 72B).
- **Fine-tuning:** Whether it's a base model or tuned for a specific task (e.g., "Instruct").

Number of parameters (billion parameters)

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In mathematical terms, the parameters (checkpoint) are a real vector:

$$\theta = (\theta_1, \theta_2, \dots, \theta_D) \in \mathbb{R}^D.$$

A "7B model" means the dimension D of this vector is approximately 7 billion.

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Generally, a higher parameter count increases expressive power but also computational cost.

Numerical representation

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- **Quantization**: Using even fewer bits (e.g., **int8**, **int4**). Drastically reduces size and speeds up inference, often with a small accuracy drop.

Precise definition of floating-point numbers

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Table 1: Bit Allocation for Floating-Point Formats

Format	Total Bits	Sign	Exponent	Mantissa	Bias
fp32	32	1	8	23	127
fp16	16	1	5	10	15
bf16	16	1	8	7	127

Comparison of fp16 and bf16

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Example (A number representable in bf16 but not in fp16)

Large number: Consider $V = 100,000$.

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Large number: Consider $V = 100,000$.

- **fp16:** The maximum representable value is 65,504. Therefore, 100,000 **overflows** to infinity (∞).
- **bf16:** Has the same dynamic range as fp32. It can be **represented without issue**.

Comparison of fp16 and bf16

Example (A number representable in fp16 but not in bf16)

A number requiring precision: Consider $V = 1 + 2^{-8} = 1.00390625$. In binary, this is $(1.00000001)_2$.

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- **fp16:** The mantissa has 10 bits. It can **accurately represent** the '1' in the 8th decimal place.
- **bf16:** The mantissa has only 7 bits. The 8th bit does not fit and is rounded. It **cannot distinguish** between 1 and $1 + 2^{-8}$.

Instruction finetuning

Suffixes like “Instruct” or “Chat” suggest the model has undergone **Instruction Finetuning**.

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- A base large language model (foundation model) is trained to predict the next word.
- Additional fine-tuning is performed using a dataset of "instruction (prompt) and desired response" pairs [28].
- This teaches the model to be a helpful assistant, follow diverse instructions, and engage in conversation.

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- Many "open models" release their **architecture and checkpoints**, but the **training data and learning algorithms** are often kept private.
- Even if a model is "public," whether **third parties can freely use it** is determined by its **license**.
- In practice, one must understand the nature of licenses like **MIT / BSD-3 / Apache 2.0 / GPL-3** and always check the **LICENSE and model card for each model**.

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Next time

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- The **neural network part** (architecture + checkpoint).
- The **surrounding components** (tokenizer, pre/post-processing, inference runtime).

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